Marketing Effectiveness & Media Budget Optimisation

An End-to-End Marketing Mix Modelling and Econometric Analysis

PostgreSQL | Python | Pandas | Statsmodels | SciPy

156	5	2023-2025
weekly observations	media channels	analysis period

Portfolio analytics project

October 2026

Data provenance note

This project uses a synthetic, business-realistic dataset created for educational and portfolio purposes. All findings in this report are model outputs from that synthetic data and do not represent the performance of any real company.

Executive Summary

This project evaluates how weekly sales are influenced by pricing, promotions, distribution, seasonal demand, competitor activity and five paid-media channels: TV, Search, Meta, YouTube and out-of-home (OOH). The objective was not only to explain historical sales, but also to estimate incremental media contribution and identify a more efficient allocation of the existing media budget.

The analysis followed an end-to-end workflow. Raw weekly data were imported and quality-checked in PostgreSQL, cleaned with an auditable set of business rules, explored in Python, and modelled using multiple regression with trend and seasonal controls. Media variables were then transformed using adstock to capture carryover and saturation to capture diminishing returns. A constrained attribution model was used to avoid implausible negative paid-media contributions, followed by counterfactual contribution analysis, ROAS estimation, response curves and constrained budget optimisation.

Metric	Result	Metric	Result
Average weekly sales	114,125 units	Average weekly media spend	BDT 15.63M
Final model R^2	0.849	Final model RMSE	5,341 units
Baseline share	70.6%	Measured-media share	29.4%
Overall historical ROAS	1.18	Optimised scenario uplift	11.5%
Annualised extra units	203,641	Annualised revenue scenario	BDT 111.9M

Executive takeaway

The model suggests that the historical media mix is not fully efficient. Under a conservative +/-30% reallocation guardrail and no increase in total media spend, the response-curve optimisation produces an estimated 11.5% uplift in media-driven sales. This is a scenario estimate, not a guaranteed causal outcome, and should be validated through staged testing.

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1. Business Context and Project Objectives

A consumer brand invests across multiple paid-media channels while weekly sales are simultaneously influenced by pricing, promotions, distribution, holidays, competitor activity and underlying market trends. Management therefore needs more than a dashboard of spend and sales: it needs a structured method for separating these effects and converting historical data into budget decisions.

The central business problem was to estimate the contribution and efficiency of each channel and then determine whether the same total media budget could be reallocated to improve expected sales response.

1.1 Core business questions

1. What are the main drivers of weekly sales?
2. How much incremental sales does each measured media channel generate?
3. Which channels appear most and least efficient historically?
4. How long can advertising effects persist after media spend occurs?
5. Which channels show signs of diminishing returns?
6. How much sales is attributable to baseline factors versus measured paid media?
7. If total media spend remains fixed, how should the budget be reallocated?
8. What assumptions and uncertainties should accompany the recommendation?

1.2 Analytical workflow

End-to-end workflow
Business understanding -> SQL validation -> data cleaning -> exploratory analysis -> econometric base model -> media transformations -> model diagnostics -> counterfactual attribution -> ROAS -> response curves -> constrained optimisation -> management recommendations.

2. Data, Variables and Analytical Tools

The analysis uses 156 clean weekly observations covering 2023-01-02 to 2025-12-22. One row represents one week at the national market level. The primary dependent variable is weekly unit sales, while revenue is retained for commercial interpretation.

2.1 Variable groups

Group	Variables	Purpose
Outcome	Sales units; Revenue	Measure business performance
Commercial controls	Average price; Promotion; Distribution	Separate non-media sales drivers
Context controls	Holiday index; Competitor activity	Capture external demand conditions
Paid media	TV; Search; Meta; YouTube; OOH spend	Estimate incremental media contribution
Time controls	Trend; annual Fourier terms	Control long-term movement and seasonality

2.2 Technology stack

- PostgreSQL: raw-data storage, structured validation, duplicate checks, range checks, temporal continuity tests and clean-table construction.
- Python/Pandas/NumPy: exploratory analysis, feature engineering and data manipulation.
- Statsmodels: OLS estimation and regression diagnostics.
- SciPy: non-negative constrained least squares and SLSQP budget optimisation.
- Matplotlib: analytical visualisation and report figures.
- Excel workbook: data dictionary, raw data inspection and business-question reference.

Security practice
Database credentials are intentionally omitted from this report. Production or portfolio code should load secrets from environment variables rather than storing passwords inside notebooks.

3. Data Quality Validation and Cleaning

The raw file was treated as an immutable source layer. The analysis first identified problems, documented them, and only then created a separate clean analytical table. This preserved traceability and avoided silently overwriting source data.

3.1 Validation framework

Check	Purpose	Outcome
Row and date range	Confirm completeness of delivery	157 raw rows spanning 2023-2025
Unique week IDs	Detect duplicate weekly records	One exact duplicate week (W130)
Missing values	Identify incomplete fields	Three isolated missing observations
Media ranges	Detect impossible spend	One negative OOH value (-0.80M)
Business rules	Detect implausible but technically valid values	One 75% promotion anomaly
Weekly continuity	Detect missing periods	No missing weekly intervals after deduplication
Revenue reconciliation	Cross-check sales x price against revenue	Consistent subject to rounding
Category consistency	Validate market labels	Single consistent national category

3.2 Cleaning decisions and audit trail

Week	Field	Raw issue	Treatment	Reason
W023	Search spend	Missing	2.56M via local interpolation	Isolated weekly gap
W068	Distribution	Missing	88.25% via local interpolation	Smooth business variable; adjacent weeks available
W119	Meta spend	Missing	3.15M via local interpolation	Isolated weekly gap
W045	OOH spend	-0.80M	3.01M via local interpolation	Negative spend invalid under project rules
W092	Promotion	75%	Reset to 0% after local review	Inconsistent with surrounding campaign pattern
W130	Entire row	Exact duplicate	Retain one record	Identical duplicate observation

After cleaning, the analytical table contained 156 unique weekly observations with no remaining missing values across model variables. Database constraints were then added to prevent invalid future records, including primary/unique keys, non-negative sales and spend rules, valid promotion and distribution ranges, and a 0-1 range for the holiday index.

Key methodological principle

A missing value is not the same as a true zero. The project therefore investigated the meaning of each missing observation before choosing an imputation rule.

4. Exploratory Data Analysis

Exploratory analysis was used to understand scale, time patterns and plausible relationships before formal modelling. The purpose was to generate hypotheses, not to claim causality.

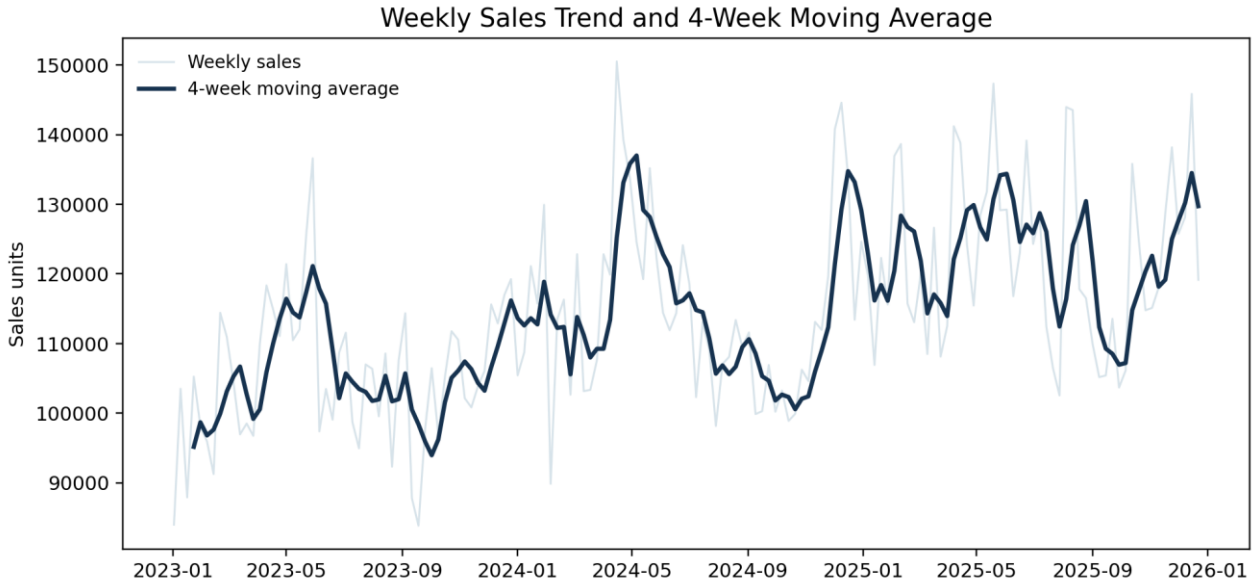


Figure 1. Weekly sales and four-week moving average.

Weekly sales display both long-term growth and recurring fluctuations. Average weekly sales increased from approximately 105,292 units in 2023 to 114,816 in 2024 and 122,414 in 2025, while average weekly measured media spend declined slightly over the same period. This immediately suggested that sales growth could not be explained by media spending alone and motivated explicit trend and seasonal controls.

Year	Avg weekly sales	Total revenue (BDT M)	Avg weekly media spend (BDT M)
2023	105,292	3,031.12	16.01
2024	114,816	3,327.36	15.61
2025	122,414	3,400.94	15.26

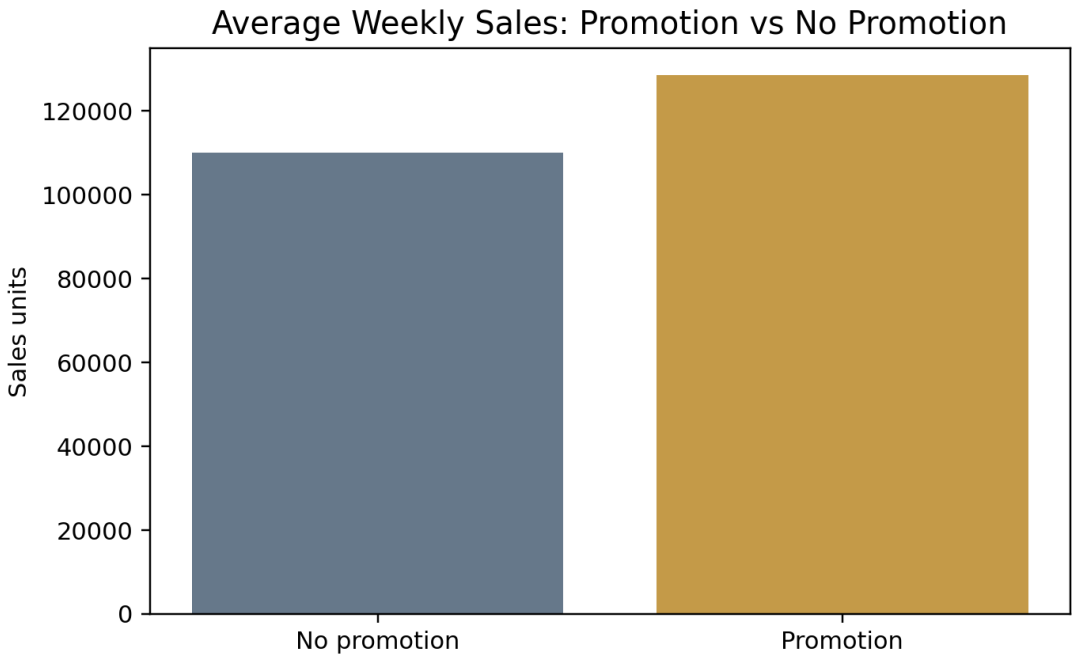


Figure 2. Average sales in promotional and non-promotional weeks.

Promotional weeks averaged approximately 128,565 units compared with 110,101 units in non-promotional weeks. This is a strong descriptive association, but it is not an incremental promotion effect because promotions may coincide with holidays, pricing changes or heavier media activity.

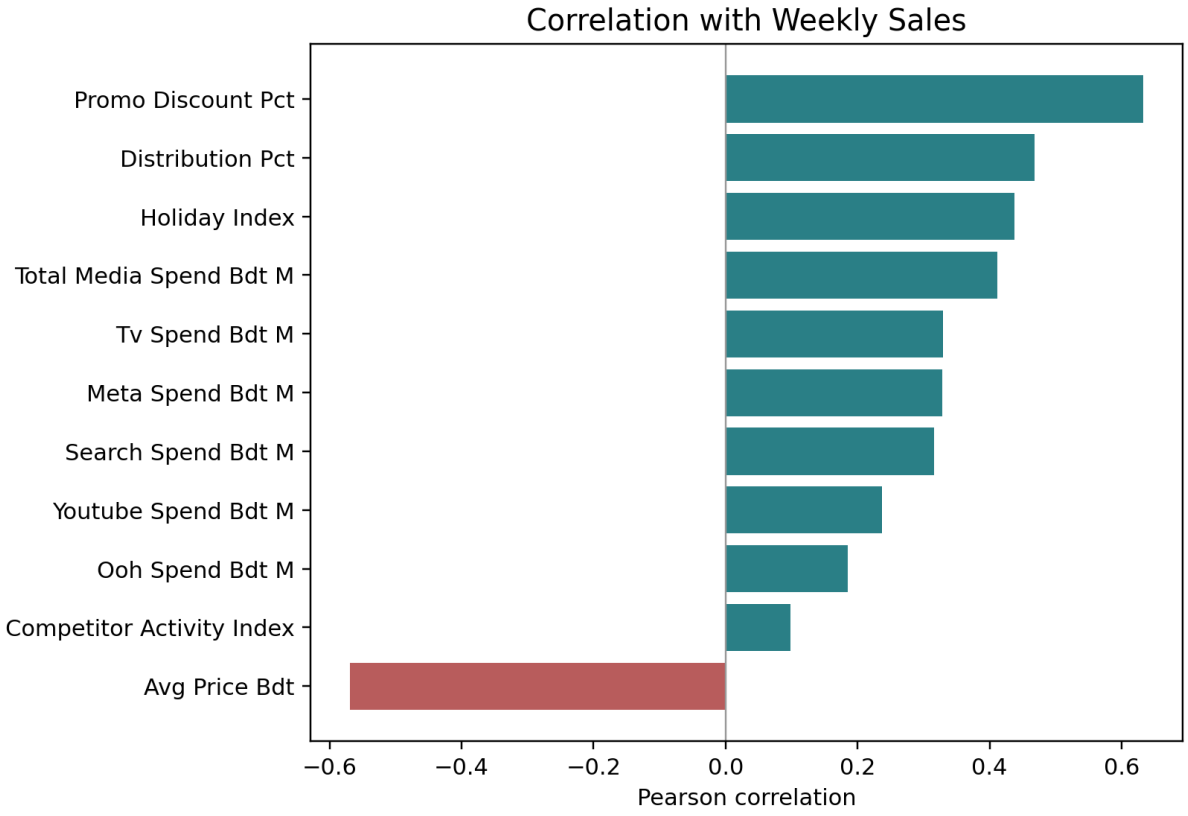


Figure 3. Bivariate correlations between weekly sales and candidate drivers.

The strongest simple relationships included promotion (+0.63), price (-0.57), distribution (+0.47), holiday activity (+0.44) and total media spend (+0.41). Because these variables can move together, correlation was treated as diagnostic evidence rather than causal attribution.

5. Econometric Framework and Base Regression

The first formal model used multiple ordinary least squares (OLS) regression. Weekly sales were modelled as a function of commercial controls, external context, a linear time trend and Fourier terms for annual seasonality.

$$Sales_t = beta_0 + beta_1\ Price_t + beta_2\ Promo_t + beta_3\ Distribution_t + beta_4\ Holiday_t + beta_5\ Competitor_t + Trend_t + Seasonality_t + epsilon_t$$

Each beta estimates a partial association holding the other included variables constant.

Equation 1. Base econometric specification

5.1 Base model results

Variable	Approx. coefficient	p-value	Interpretation
Average price	-367	0.004	Higher realised price is associated with lower unit demand
Promotion	-67	0.889	Not separately identified once price and other controls are included
Distribution	+1,755	<0.001	A 1-point increase in distribution is associated with higher weekly sales
Holiday index	+10,960	<0.001	Holiday intensity is strongly associated with higher demand
Competitor activity	-179	0.003	Higher competitor activity is associated with lower sales
Trend	+39	0.058	Positive underlying growth, borderline at conventional 5% level
Annual sine term	+8,769	<0.001	Strong recurring annual seasonality

The base model achieved $R^2 = 0.816$ and adjusted $R^2 = 0.804$. The price-promotion relationship was highly collinear: their VIF values were approximately 32.5. This explains why promotion appeared strong in simple EDA but became statistically unstable once realised price entered the regression.

5.2 Why raw media spend was not enough

Adding current-week media spend improved adjusted R^2 from 0.804 to 0.815, but individual channel coefficients remained statistically uncertain. This was expected because a same-week linear model assumes that advertising has no carryover and that every additional unit of spend generates a constant response. Both assumptions are commercially unrealistic.

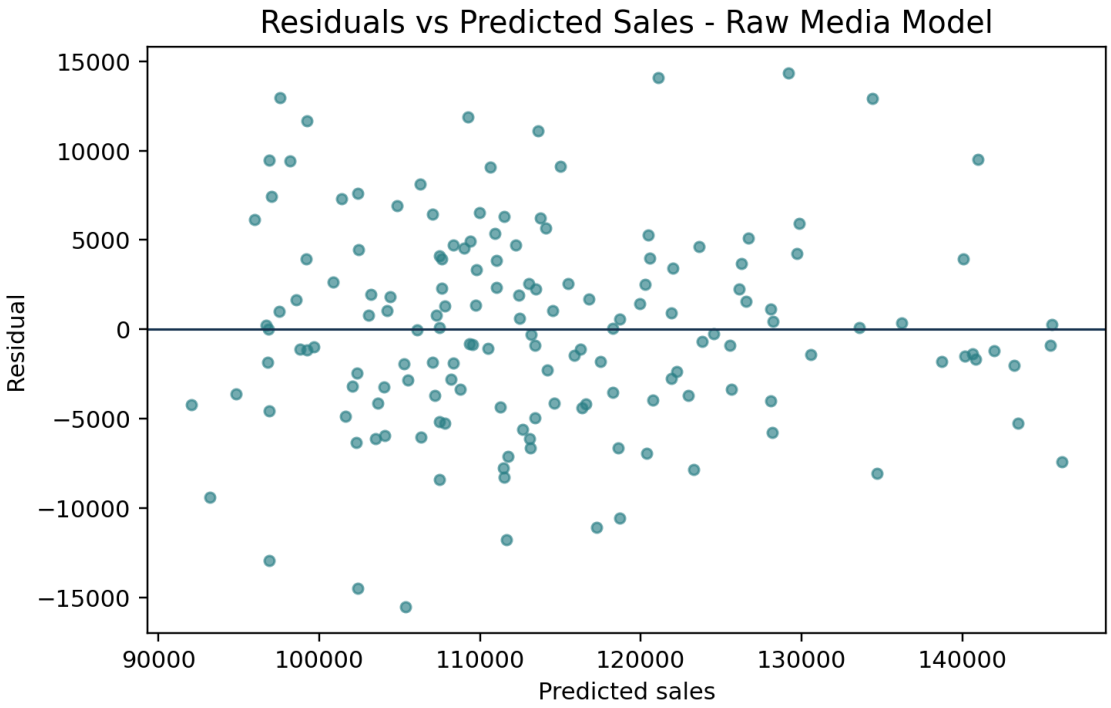


Figure 4. Residual pattern for the raw-media regression model.

The raw-media model produced a Durbin-Watson statistic of 1.84, close to the ideal value of 2 for no strong first-order serial correlation. The Breusch-Pagan p-value was 0.981, providing no strong evidence of heteroscedasticity under this specification.

6. Marketing Mix Modelling: Adstock and Saturation

To represent how advertising works more realistically, media spend was transformed in two stages. Adstock represented persistence through time, while saturation represented diminishing marginal returns at higher media levels.

6.1 Adstock: modelling carryover

$$Adstock_t = Spend_t + lambda \times Adstock_(t\text{-}1)$$

The decay parameter lambda determines how much of the previous media stock carries into the next period.

Equation 2. Geometric adstock

Channel	Decay parameter	Interpretation
TV	0.60	Longer carryover
Search	0.20	More immediate effect
Meta	0.35	Moderate-short carryover
YouTube	0.45	Moderate carryover
OOH	0.50	Moderate-long carryover

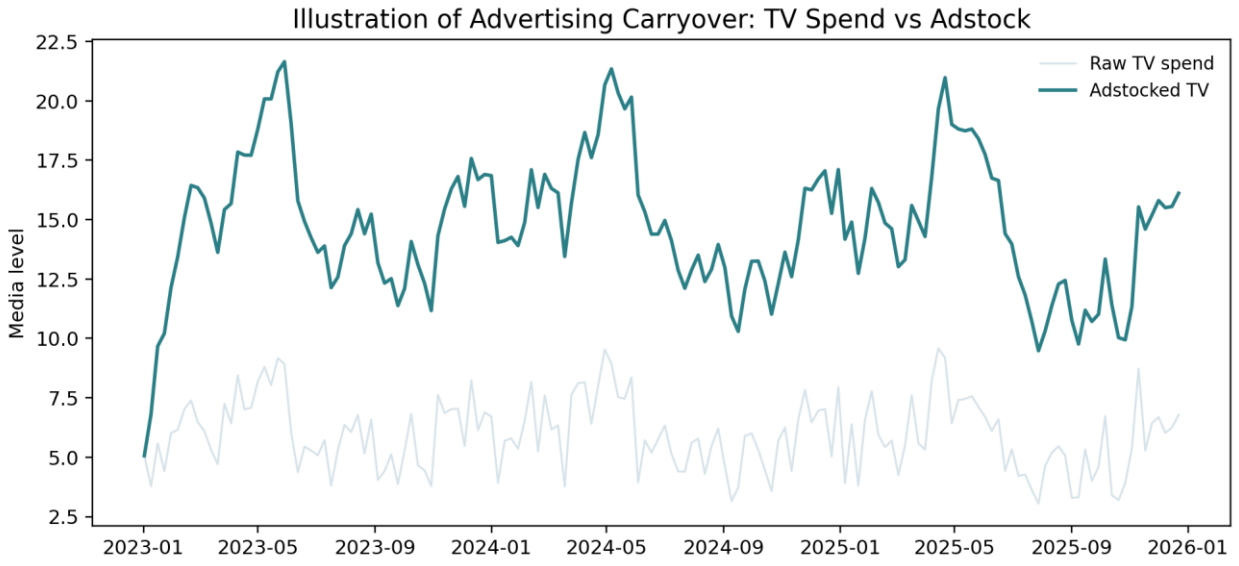


Figure 5. Example of raw TV spend transformed into a smoother adstock series.

6.2 Saturation: modelling diminishing returns

$$Saturation(x) = x / (x + theta)$$

theta is the transformed media level at which the response reaches 50% of its asymptotic maximum under this simple form.

Equation 3. Saturation transformation

Channel	Theta
TV	14.0
Search	5.5
Meta	7.5
YouTube	5.0
OOH	6.5

Interpretation

Adstock answers when advertising can continue to matter; saturation answers how the incremental response changes as spend increases. Together they produce a more plausible media-response structure than a same-week linear coefficient.

7. Model Validation and Selection

Four specifications were compared on the same 156-week sample. Fit was assessed using R^2, adjusted R^2, AIC and RMSE, while business plausibility and diagnostics were also considered.

Model	R^2	Adjusted R^2	AIC	RMSE
Base	0.816	0.804	3173.5	5,896
Raw Media	0.833	0.815	3168.9	5,627
Adstock	0.846	0.830	3155.5	5,391
Adstock + Saturation	0.849	0.833	3152.5	5,339

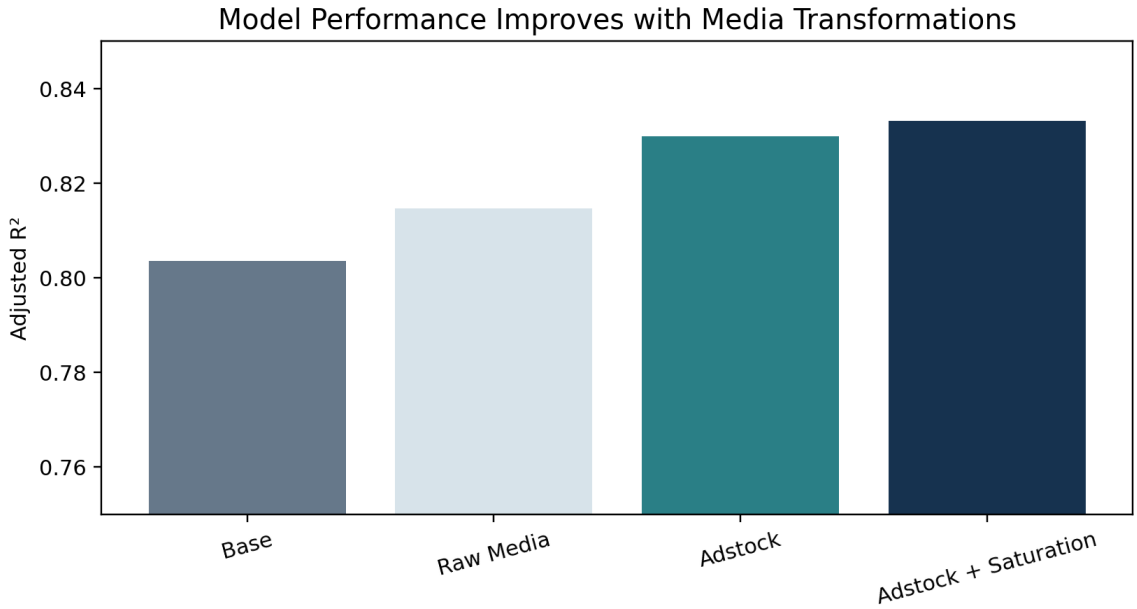


Figure 6. Adjusted R^2 across successive model specifications.

The final adstock-plus-saturation specification improved adjusted R^2 to 0.833, reduced RMSE to approximately 5,339 units, and achieved the lowest AIC. The OLS version produced positive significant coefficients for YouTube and OOH, while Meta was negative and statistically uncertain. For commercial attribution, a non-negative paid-media constraint was therefore imposed while leaving non-media controls unconstrained.

The constrained attribution model retained essentially the same predictive performance: R^2 = 0.849 and RMSE = 5,341 units. Under the constraint, Meta's coefficient was approximately zero rather than negative. This was interpreted as insufficient independent evidence to isolate a positive Meta effect, not proof that Meta has no value.

8. Incremental Contribution, Sales Effects and ROAS

Because the final media variables are nonlinear transformations, raw regression coefficients are not interpreted as units per BDT million of spend. Instead, channel contribution was estimated using counterfactual predictions.

$$\text{Contribution_channel} = \text{Predicted Sales_actual} - \text{Predicted Sales_without channel}$$

All other model inputs are held constant while the selected channel effect is set to zero.

Equation 4. Counterfactual channel contribution

Across the full period, the model predicted approximately 17.80 million units. Of this, 12.57 million (70.6%) were classified as baseline/non-media demand and 5.23 million (29.4%) as measured-media contribution.

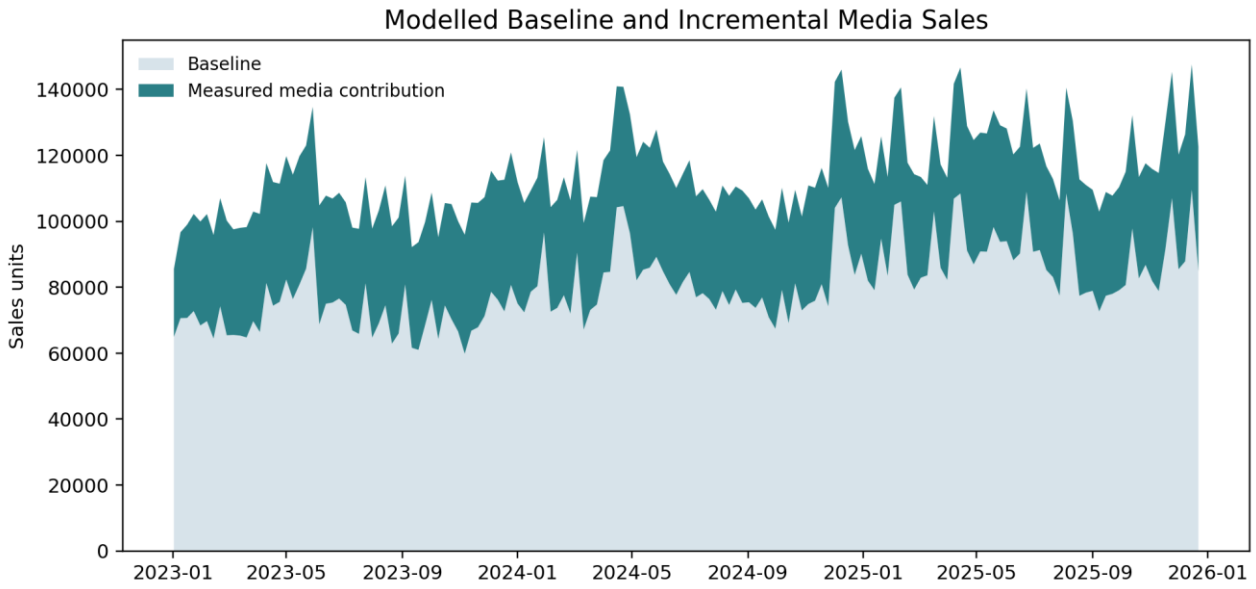


Figure 7. Modelled baseline sales and measured-media contribution over time.

Channel	Spend (BDT M)	Incremental units	Contribution share	Spend share
Tv	928.32	934,004	17.8%	38.1%
Search	369.25	425,338	8.1%	15.1%
Meta	492.38	0	0.0%	20.2%
Youtube	301.97	1,697,888	32.4%	12.4%
Ooh	346.13	2,176,502	41.6%	14.2%

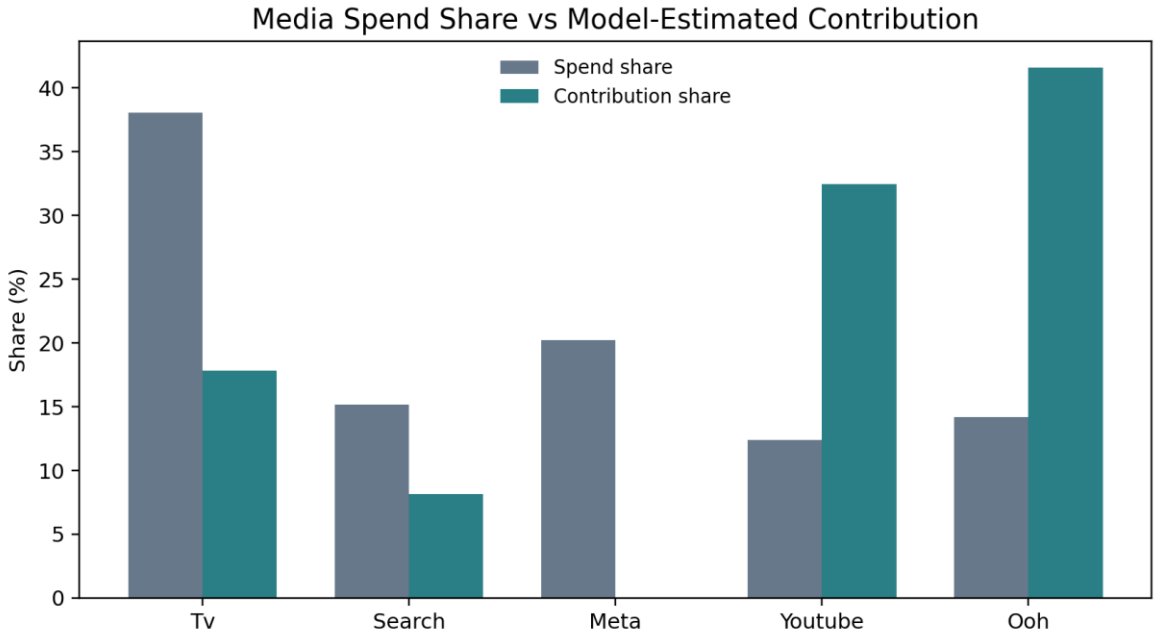


Figure 8. Historical spend share compared with model-estimated contribution share.

8.1 ROAS

$$\text{ROAS} = \text{Incremental Revenue} / \text{Media Spend}$$

Incremental revenue is calculated from model-estimated incremental units multiplied by weekly realised price.

Equation 5. Return on advertising spend

Channel	Incremental revenue (BDT M)	Historical ROAS	Incremental units / BDT M
Tv	513.4	0.55	1,006
Search	233.7	0.63	1,152
Meta	0.0	0.00	0
Youtube	933.1	3.09	5,623
Ooh	1,195.8	3.45	6,288

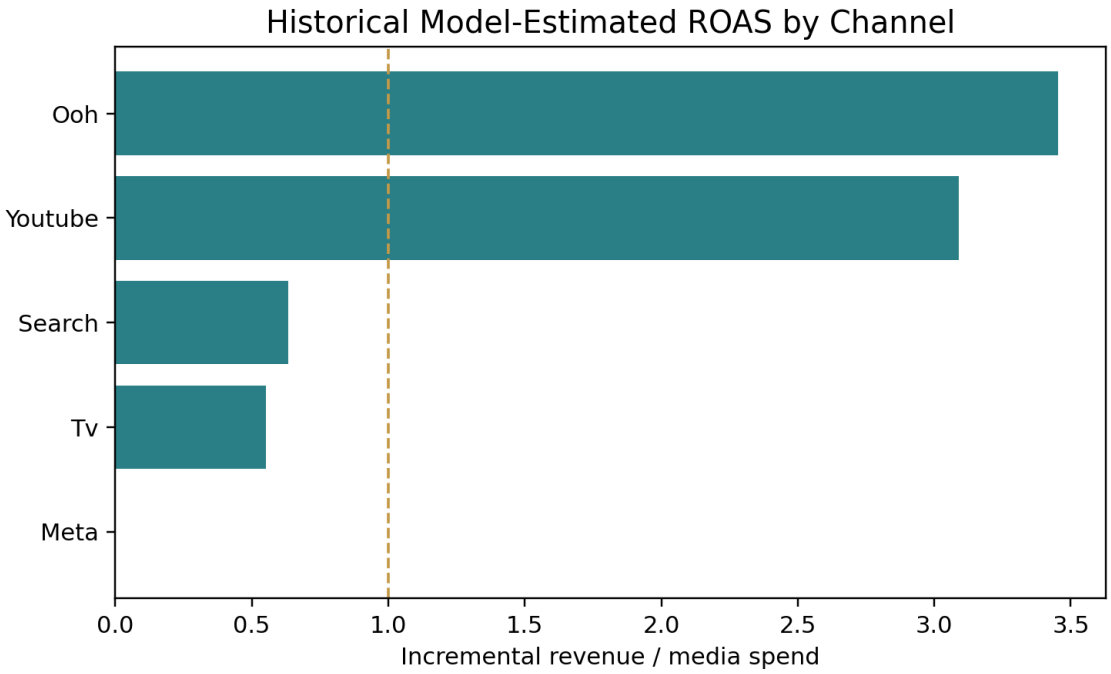


Figure 9. Historical model-estimated ROAS by channel. The dashed line marks ROAS = 1.

Important distinction

The dataset supports ROAS because revenue is observed. A defensible profit-based ROI would additionally require product margin, cost of goods sold and relevant variable costs. Any margin-based ROI shown during training should be labelled as an assumption rather than observed business data.

9. Response Curves and Budget Optimisation

Historical ROAS is an average return and should not be extrapolated indefinitely. Budget decisions depend on marginal return: the expected response to the next unit of spend at the current position on a channel response curve.

$$\text{Response}(s) = \text{beta} \times s \,/\, [s + \text{theta}(1 - \text{lambda})]$$

This simplified planning form assumes a stable weekly spend long enough for geometric adstock to approach steady state.

Equation 6. Steady-state planning response

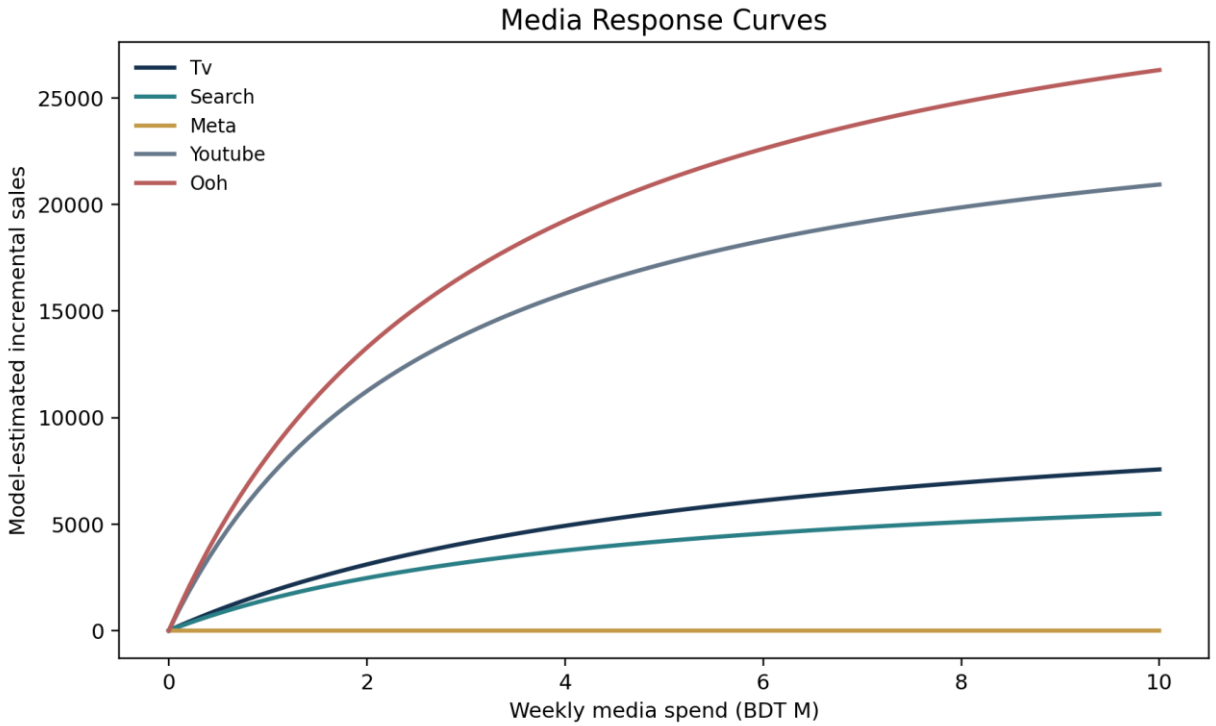


Figure 10. Estimated media response curves used for planning scenarios.

9.1 Marginal returns at the current allocation

Channel	Current weekly spend (BDT M)	Marginal units per BDT M	Marginal ROAS
Tv	5.95	495	0.27
Search	2.37	758	0.42
Meta	3.16	0	0.00
Youtube	1.94	3,344	1.84
Ooh	2.22	3,790	2.08

At the observed average weekly allocation, OOH and YouTube showed the strongest model-estimated marginal opportunity. TV and Search had positive but lower marginal return, while Meta remained unidentified under the constrained attribution model.

9.2 Constrained optimisation

The total average weekly budget was held fixed at approximately BDT 15.63M. To avoid an unrealistic mathematical solution, each channel was limited to a maximum +/-30% change from its historical average allocation. The SLSQP optimiser then maximised total steady-state media response within these constraints.

Channel	Current spend	Optimised spend	Change (BDT M)	Change
Tv	5.95	4.94	-1.01	-17.0%
Search	2.37	3.08	+0.71	+30.0%
Meta	3.16	2.21	-0.95	-30.0%
Youtube	1.94	2.52	+0.58	+30.0%
Ooh	2.22	2.88	+0.67	+30.0%

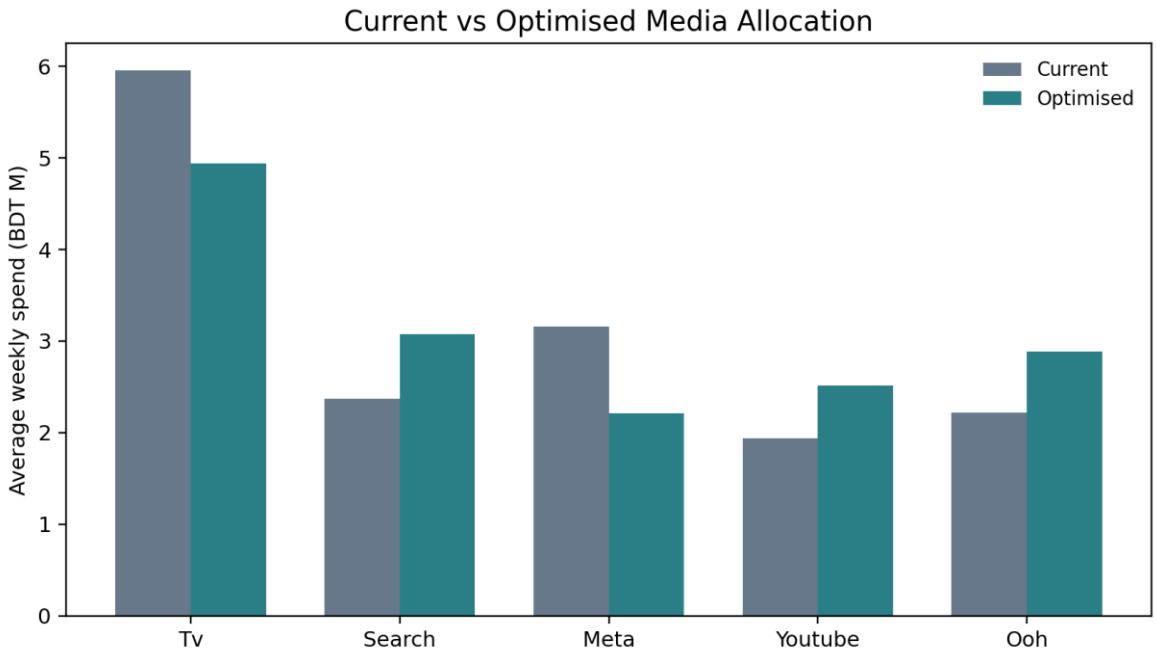


Figure 11. Current and optimised weekly media allocation under +/-30% channel guardrails.

The constrained scenario increased model-estimated steady-state media response from approximately 34,019 to 37,935 incremental units per week, an uplift of 11.5% without increasing the total media budget. Annualised mechanically across 52 weeks, this corresponds to approximately 203,641 extra units and BDT 111.9M of incremental revenue at the historical average price. These figures are scenario estimates rather than guarantees.

9.3 Sensitivity analysis

Maximum allowed channel change	Estimated media-response uplift
+/-15%	6.2%
+/-30%	11.5%
+/-50%	17.6%

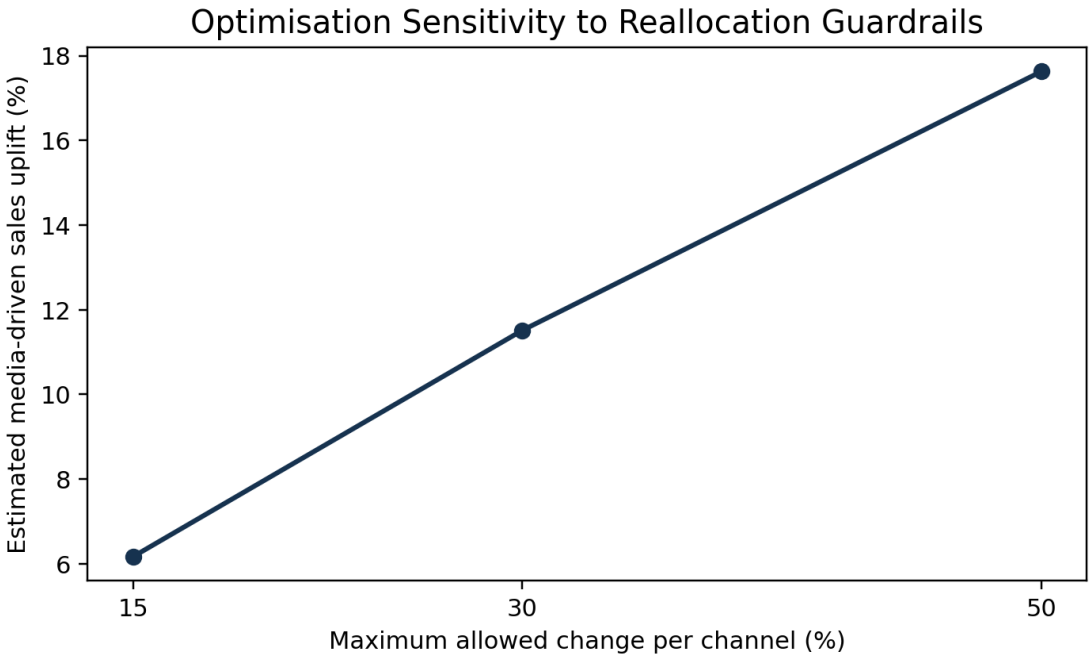


Figure 12. Sensitivity of estimated uplift to budget-reallocation guardrails.

Larger reallocation freedom produces higher modelled uplift, but it also moves further from historical behaviour and increases implementation and model risk. The +/-30% scenario was therefore selected as a balanced planning case rather than the unconstrained mathematical optimum.

10. Business Recommendations

10.1 Rebalance rather than increase the total budget

Use the existing budget more efficiently before seeking incremental investment. The model supports a gradual reduction in relatively saturated or weakly identified allocations and a controlled increase in channels with stronger marginal response, particularly OOH and YouTube in the current specification.

10.2 Validate uncertain channels with experiments

Meta could not be independently attributed with confidence because its media pattern overlaps with other channels. Rather than interpreting the constrained zero contribution literally, management should use geo tests, holdouts, incrementality experiments or campaign-level variation to obtain independent evidence and calibrate the next model refresh.

10.3 Refresh the model on a regular cadence

Media effectiveness, pricing, competitive behaviour and channel saturation change over time. The model should be re-estimated as new weekly data accumulate and should be monitored for coefficient stability, changes in response curves and shifts in optimal budget allocation.

10.4 Implement through staged budget changes

The optimisation result is best treated as a decision-support scenario. A prudent implementation would move budgets in stages, monitor realised incremental performance, and feed the observed results back into model calibration rather than implementing the full reallocation immediately.

Recommended management message

The analysis does not say that one channel is universally "good" and another is "bad". It says that, at the historical allocation and under the fitted response curves, the marginal opportunity differs by channel. Budget should therefore be rebalanced using incremental opportunity, not historical spend or average ROAS alone.

11. Assumptions, Limitations and Risk

Area	Limitation / risk	Implication
Data provenance	Synthetic portfolio dataset	Results illustrate methodology rather than a real market outcome
Causal identification	Observational time-series data	Regression associations are not equivalent to experimental causal effects
Media correlation	Channels often move together	Individual channel attribution can be unstable or weakly identified
Adstock/saturation	Parameters are modelling assumptions	Should be tuned and validated using holdout fit, business logic and experiments
Omitted variables	No weather, brand equity, creative quality, inventory or macro variables	Unobserved drivers may bias coefficients
Optimisation	Steady-state weekly spend assumption	Does not optimise campaign timing across weeks
Profitability	No margin or COGS data	ROAS is supported; true profit ROI is not directly observable
Forecast risk	Response curves are estimated from historical ranges	Large extrapolations beyond observed spend are unreliable

These limitations do not invalidate the analysis; they define how the outputs should be interpreted and what evidence should be collected next. A mature analytics process makes model uncertainty explicit rather than presenting point estimates as guaranteed outcomes.

12. Conclusion

This project demonstrates a complete marketing analytics workflow that moves from raw operational data to a management decision. The analysis first established data integrity, then separated important commercial and seasonal sales drivers, represented media carryover and saturation, quantified model-estimated channel contribution, and finally translated those effects into budget scenarios.

The final constrained model explained approximately 84.9% of observed weekly sales variation. It estimated that 29.4% of modelled sales were associated with measured-media contribution, while 70.6% came from baseline and non-media factors. Historical efficiency was strongest for OOH and YouTube under this synthetic dataset, and the +/-30% reallocation scenario produced an estimated 11.5% improvement in media-driven response with no increase in total spend.

The strongest practical lesson is that media decisions should not be based only on correlation, raw regression coefficients or historical average ROAS. A defensible decision process combines data quality, econometric controls, nonlinear media response, counterfactual attribution, marginal returns, business constraints and explicit uncertainty.

Appendix A. Data Dictionary

Variable	Meaning	Unit / type	Role
Week_Start	Start date of reporting week	Date	Time index
Week_ID	Sequential weekly identifier	Text	Audit / joins
Market	Geographic reporting level	Category	Dimension
Sales_Units	Weekly units sold	Units	Primary dependent variable
Revenue_BDT_M	Weekly sales revenue	BDT millions	Commercial KPI
Avg_Price_BDT	Average realised selling price	BDT/unit	Control variable
Promo_Discount_Pct	Average promotional discount	Percent	Control variable
Distribution_Pct	Weighted distribution / availability	Percent	Control variable
Holiday_Index	Holiday/festival intensity	0-1 index	Control variable
Competitor_Activity_Index	Relative competitor activity	Index	Control variable
TV_Spend_BDT_M	Weekly TV spend	BDT millions	Paid-media driver
Search_Spend_BDT_M	Weekly paid-search spend	BDT millions	Paid-media driver
Meta_Spend_BDT_M	Weekly social-platform spend	BDT millions	Paid-media driver
YouTube_Spend_BDT_M	Weekly video-platform spend	BDT millions	Paid-media driver
OOH_Spend_BDT_M	Weekly out-of-home spend	BDT millions	Paid-media driver

Appendix B. Key Mathematical and Statistical Concepts

Concept	Project meaning
Multiple regression	Estimates the relationship between weekly sales and multiple drivers simultaneously, holding the other included variables constant.
Coefficient	Estimated change in the dependent variable for a one-unit change in a predictor, conditional on the other included predictors.
R-squared	Proportion of observed variation in sales explained by the model; it is not a measure of causal correctness.
Adjusted R-squared	R-squared adjusted for model complexity, useful when comparing models with different numbers of predictors.
p-value	Evidence against the null hypothesis for a coefficient under the model assumptions; it is not a binary truth test.
Multicollinearity	Strong overlap among explanatory variables that can make individual coefficients unstable. Price and promotion were the clearest example.
VIF	Variance Inflation Factor, a diagnostic for multicollinearity. Values above roughly 5-10 warrant investigation rather than automatic deletion.
Residual	Actual sales minus model-predicted sales. Residual patterns reveal model misspecification.
Durbin-Watson	Diagnostic for first-order residual autocorrelation; values near 2 are generally reassuring.
Heteroscedasticity	Non-constant residual variance. The project used a Breusch-Pagan test as a diagnostic.
Adstock	A transformation representing carryover of advertising effects into later periods.
Saturation	A nonlinear transformation representing diminishing returns to additional media activity.
Counterfactual attribution	Estimates a channel contribution by comparing predicted sales under observed media with predicted sales after removing that channel effect.
Baseline sales	Modelled sales remaining without the incremental effect of measured paid media, while other controls are retained.
ROAS	Incremental revenue divided by advertising spend.
Marginal return	Expected response to the next unit of spend at the current media level.
Budget optimisation	Allocates a fixed total budget across channels to maximise model-estimated response subject to business constraints.

$$Adstock_t = Spend_t + lambda \times Adstock_(t\text{-}1)$$

Core formula A

$$Saturation(x) = x / (x + theta)$$

Core formula B

$$ROAS = Incremental\ Revenue / Media\ Spend$$

Core formula C

$$\text{Contribution_channel} = \text{Prediction_actual} - \text{Prediction_without channel}$$

Core formula D

Appendix C. Selected SQL Logic

The report focuses on representative SQL patterns rather than reproducing every line of the working scripts. The raw source table should remain unchanged; cleaning should create a separate analysis-ready table and maintain an audit trail.

C.1 Duplicate and completeness checks

```
SELECT week_id, COUNT(*) AS occurrences
FROM marketing_mix_raw
GROUP BY week_id
HAVING COUNT(*) > 1;

SELECT
  COUNT(*) - COUNT(search_spend_bdt_m) AS missing_search,
  COUNT(*) - COUNT(meta_spend_bdt_m)   AS missing_meta,
  COUNT(*) - COUNT(distribution_pct)   AS missing_distribution
FROM marketing_mix_raw;
```

C.2 Temporal continuity

```
WITH ordered_weeks AS (
  SELECT week_start,
         LAG(week_start) OVER (ORDER BY week_start) AS previous_week
  FROM marketing_mix_raw
)
SELECT previous_week, week_start, week_start - previous_week AS gap_days
FROM ordered_weeks
WHERE week_start - previous_week <> 7;
```

C.3 Business-rule validation

```
SELECT *
FROM marketing_mix_raw
WHERE promo_discount_pct < 0
   OR promo_discount_pct > 40
   OR distribution_pct < 0
   OR distribution_pct > 100
   OR tv_spend_bdt_m < 0
   OR search_spend_bdt_m < 0
   OR meta_spend_bdt_m < 0
   OR youtube_spend_bdt_m < 0
   OR ooh_spend_bdt_m < 0;
```

Appendix D. Selected Python Logic

The extracts below show the core modelling logic. Database connection credentials are deliberately excluded.

D.1 Adstock and saturation

```
def geometric_adstock(series, decay):
    output = []
    carryover = 0
    for spend in series:
        carryover = spend + decay * carryover
        output.append(carryover)
    return np.array(output)

def saturation(x, theta):
    return x / (x + theta)
```

D.2 Counterfactual attribution

```
for channel, variable in channel_transform_map.items():
    counterfactual = df.copy()
    counterfactual[variable] = 0
    predicted_without_channel = constrained_predict(counterfactual)
    df[f"{channel}_contribution"] = (
        df["predicted_sales"] - predicted_without_channel
    )
```

D.3 Budget optimisation

```
result = minimize(
    objective,
    x0=current_allocation,
    method="SLSQP",
    bounds=bounds,
    constraints=[budget_constraint],
    options={"maxiter": 2000, "ftol": 1e-12}
)
```

Portfolio reproducibility note
The original project ZIP contains the source dataset, SQL scripts, Jupyter notebooks, model outputs and allocation summary. This report intentionally presents only the informative and reproducible parts of those files and excludes temporary or redundant outputs.